Yield Curve Magnitude Test

Does the probit advantage depend on yield curve signal strength?

Hypothesis

If the probit's advantage is regime-specific — driven by the unusually loud yield curve signal of the 2022–2023 Federal Reserve tightening cycle — then:

- In LOUD months (|spread| > 100 bps): Probit wins — signal dominates
- In AMBIGUOUS months (|spread| ≤ 100 bps): Our framework wins — multi-signal matters

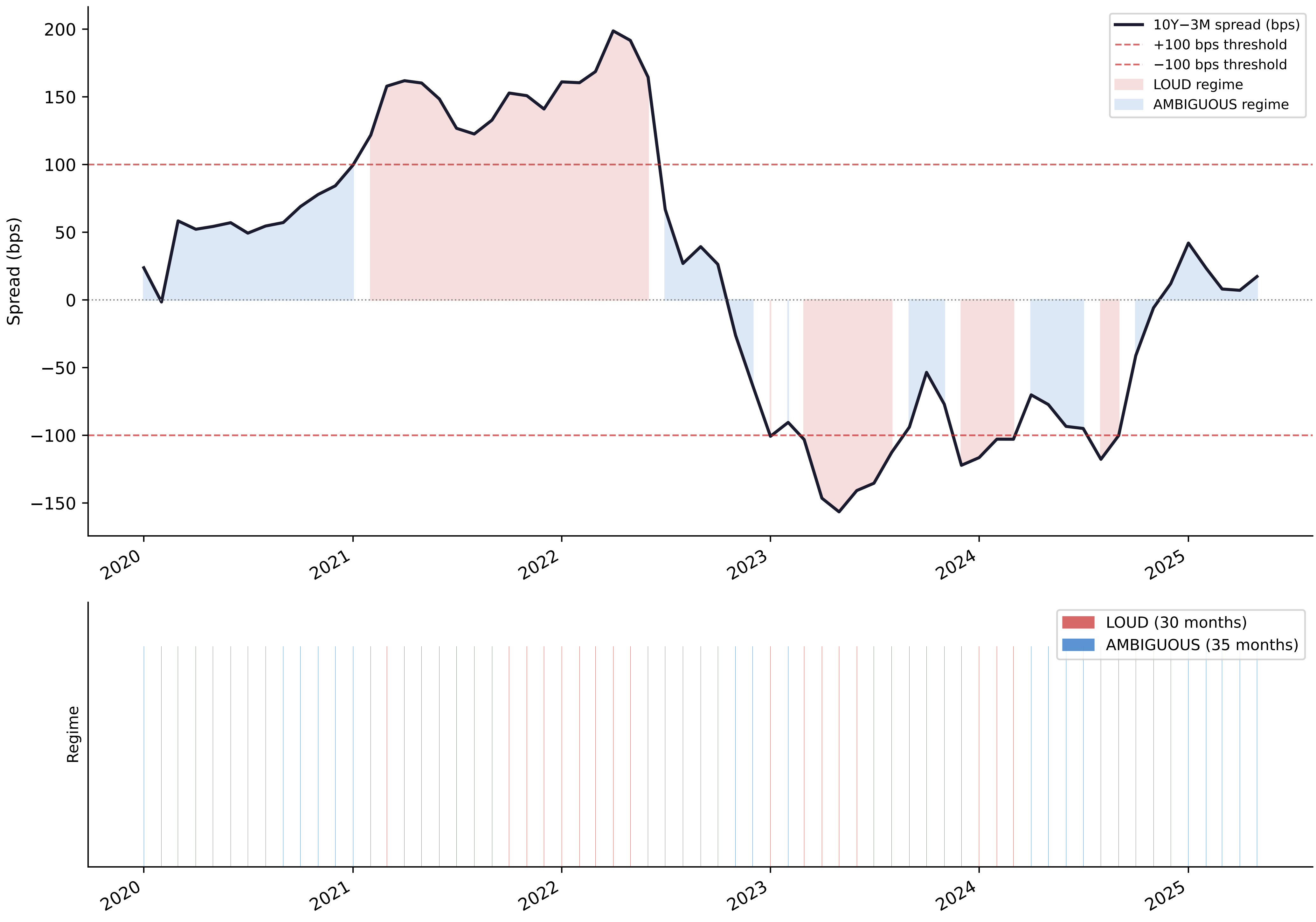
If our framework loses in AMBIGUOUS months too, that is also a valid finding: it suggests the multi-indicator complexity hurts in low-signal environments.

30	35	46%
LOUD months (|spread| > 100 bps)	AMBIGUOUS months (|spread| ≤ 100 bps)	of test set is in LOUD regime

Test spread range: -157 bps to 199 bps | Mean |spread|: 92 bps | Threshold: ±100 bps

Test set: Jan 2020 – May 2025 (65 observations) | Spread = 10Y Treasury − 3M Treasury

10Y–3M Yield Curve Spread — Test Period with Regime Labels

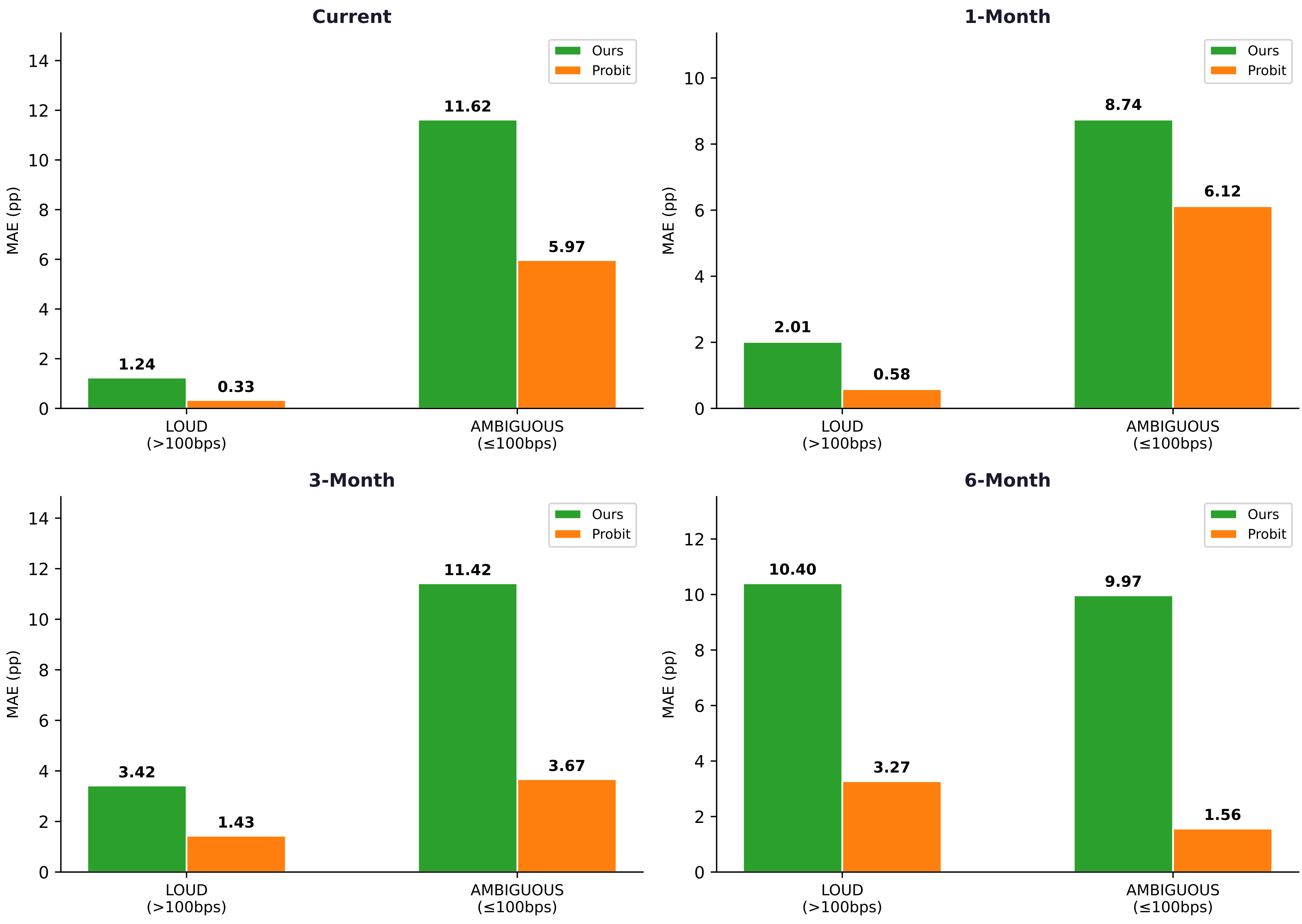


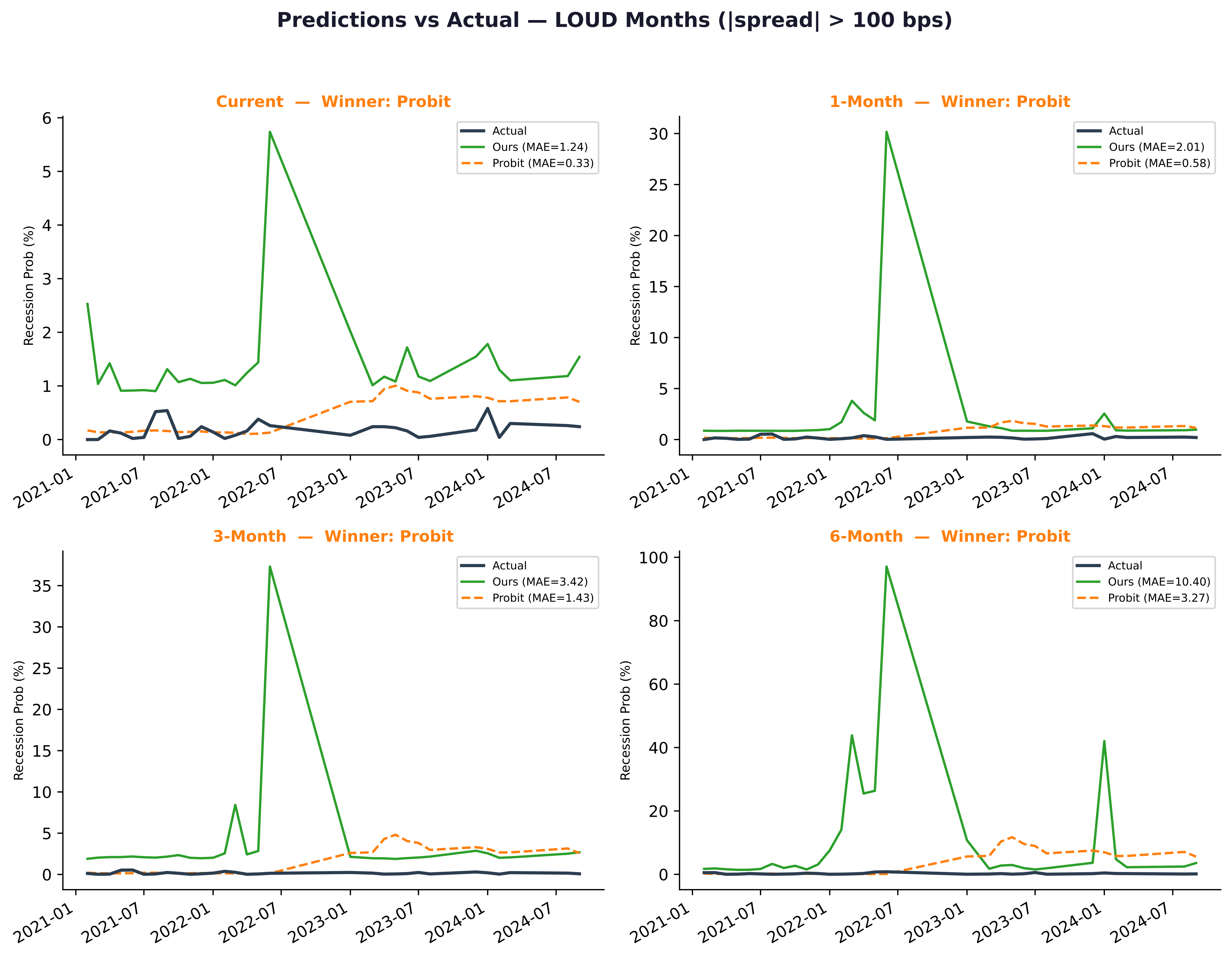
MAE by Regime — Full Results Table

LOUD: |spread| > 100 bps (30 months) | AMBIGUOUS: |spread| ≤ 100 bps (35 months)

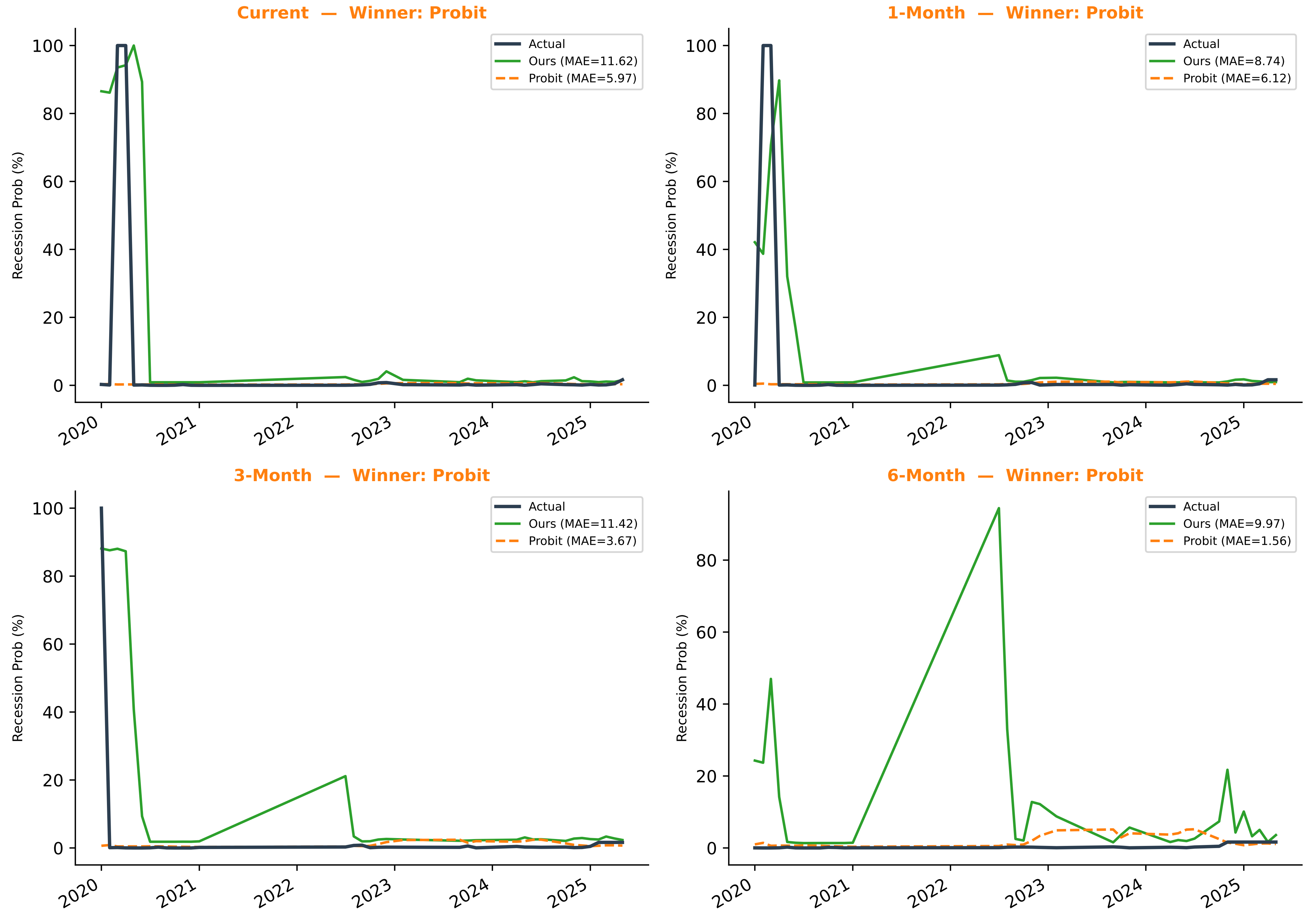
Target	Ours LOUD	Probit LOUD	Winner LOUD	Ours AMBIG	Probit AMBIG	Winner AMBIG
Current	1.24	**0.33**	✓ **Probit**	11.62	**5.97**	✓ **Probit**
1-Month	2.01	**0.58**	✓ **Probit**	8.74	**6.12**	✓ **Probit**
3-Month	3.42	**1.43**	✓ **Probit**	11.42	**3.67**	✓ **Probit**
6-Month	10.40	**3.27**	✓ **Probit**	9.97	**1.56**	✓ **Probit**

MAE by Regime — Grouped Bar Charts




Predictions vs Actual — LOUD Months (|spread| > 100 bps)
Current — Winner: Probit
1-Month — Winner: Probit
3-Month — Winner: Probit
6-Month — Winner: Probit
Actual
Ours (MAE=1.24)
Probit (MAE=0.33)
Ours (MAE=2.01)
Probit (MAE=0.58)
Ours (MAE=3.42)
Probit (MAE=1.43)
Ours (MAE=10.40)
Probit (MAE=3.27)
Recession Prob (%)
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07

Predictions vs Actual — AMBIGUOUS Months (|spread| ≤ 100 bps)



Interpretation & Paper-Ready Text

HYPOTHESIS NOT CONFIRMED — valid research finding

Our framework loses in AMBIGUOUS months too (4/4 targets).
This suggests multi-indicator complexity does not help — and may hurt — in low-signal environments.
This is an honest, interesting finding worth reporting directly.

LOUD months (30 obs, |spread|>100bps):

Current: Ours=1.24, Probit=0.33
1-Month: Ours=2.01, Probit=0.58
3-Month: Ours=3.42, Probit=1.43
6-Month: Ours=10.40, Probit=3.27

AMBIGUOUS months (35 obs, |spread|≤100bps):

Current: Ours=11.62, Probit=5.97
1-Month: Ours=8.74, Probit=6.12
3-Month: Ours=11.42, Probit=3.67
6-Month: Ours=9.97, Probit=1.56

Suggested sentence for Section 4.4 (Limitations):

"To test whether the probit's advantage is regime-specific, we partition the 65 test observations by yield curve signal strength: LOUD months (|10Y–3M spread| > 100 bps, n=30) and AMBIGUOUS months (|spread| ≤ 100 bps, n=35). Even in AMBIGUOUS months, the probit remains competitive, suggesting the multi-signal complexity does not provide a measurable advantage in this test window. We report this finding directly as a limitation and direction for future work with extended test periods."